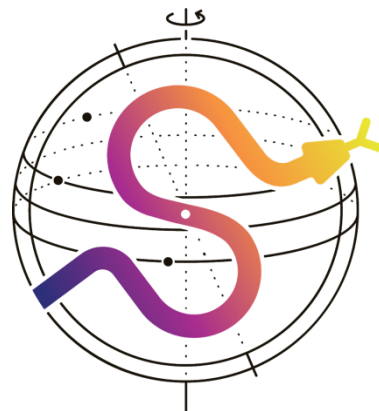




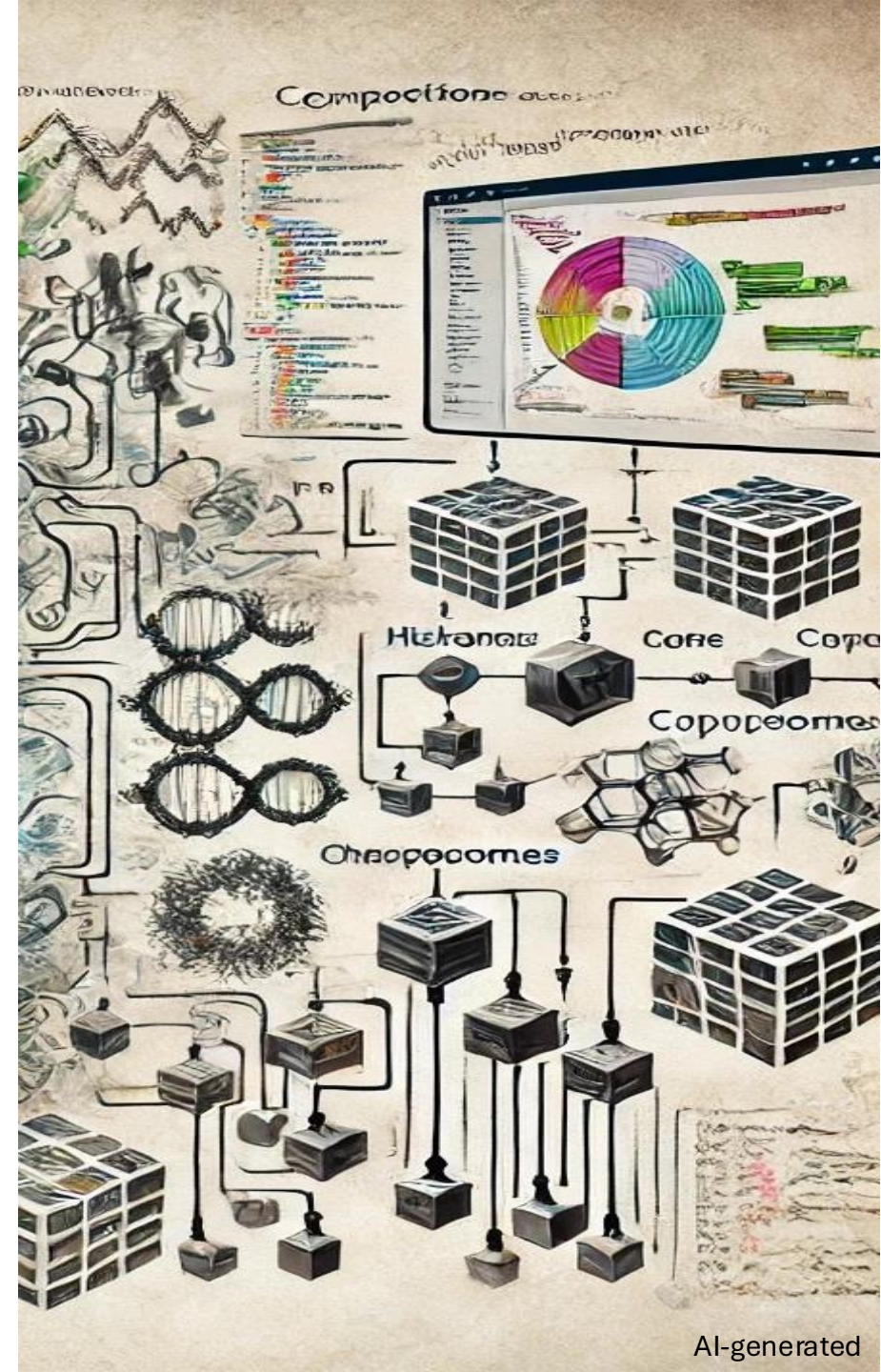
Accelerating Genomic Data Science and AI/ML with Composability

Nezar Abdennur, PhD
SciPy 2025, Tacoma, WA
7/10/2025

Composable

[kuhm-pohz-uh-buhl] adj.

Capable of being composed, or made up of smaller parts that can be combined to create something larger and more complex.



AI-generated

Bridging technologies with open standards

Target environment:



Portable

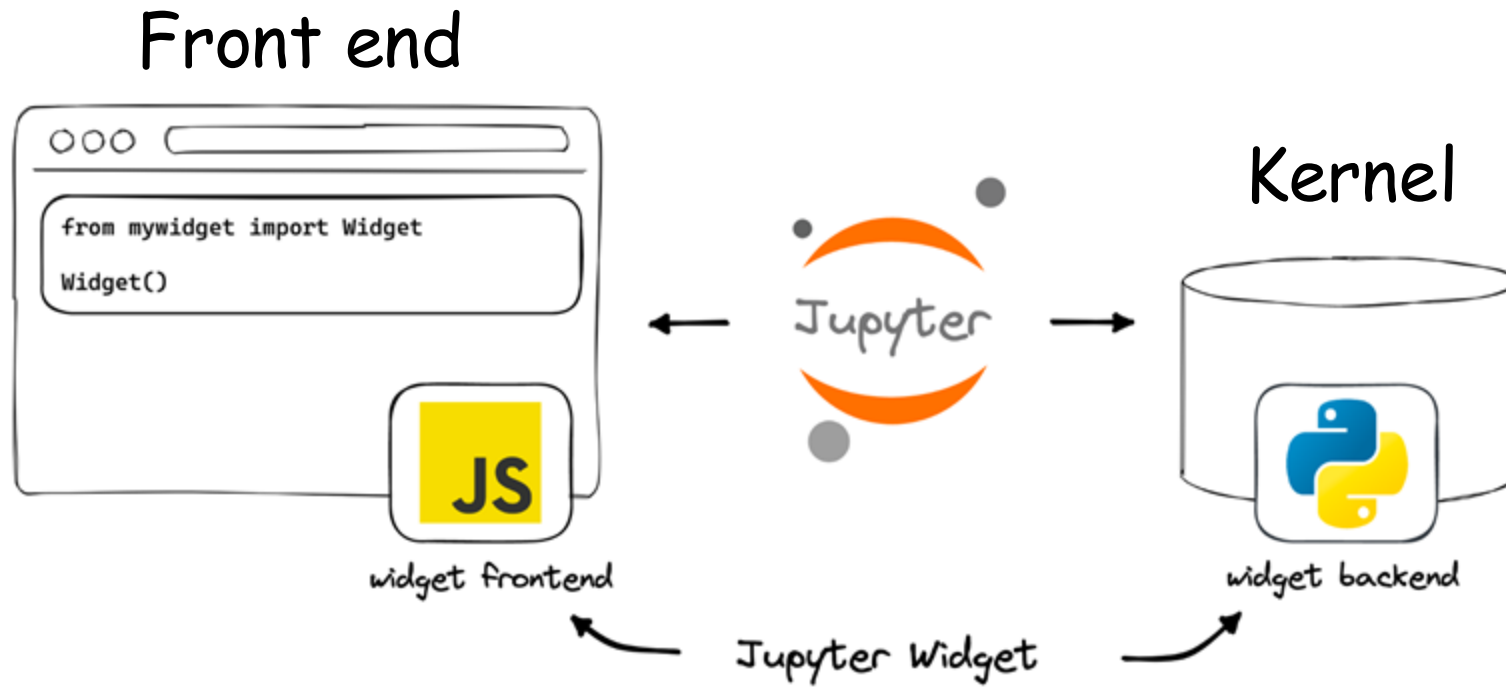
Reusable

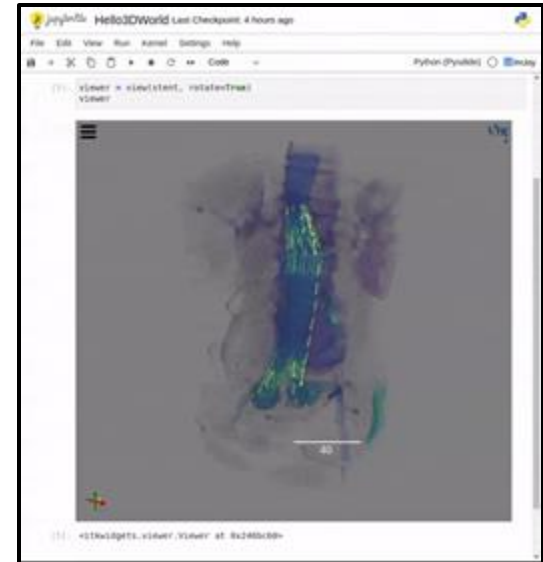
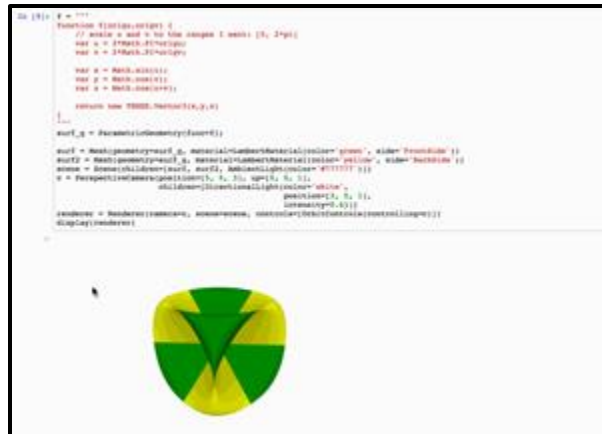
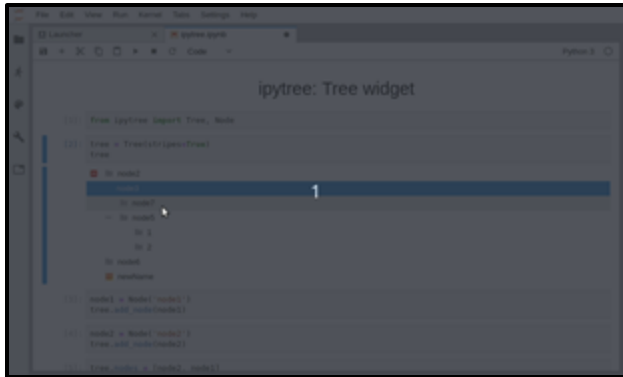
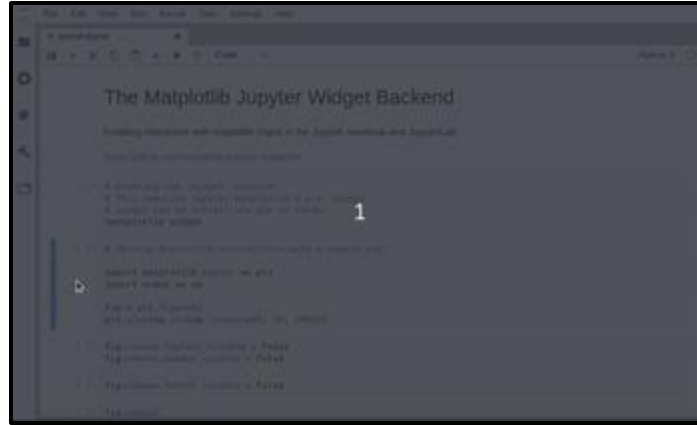
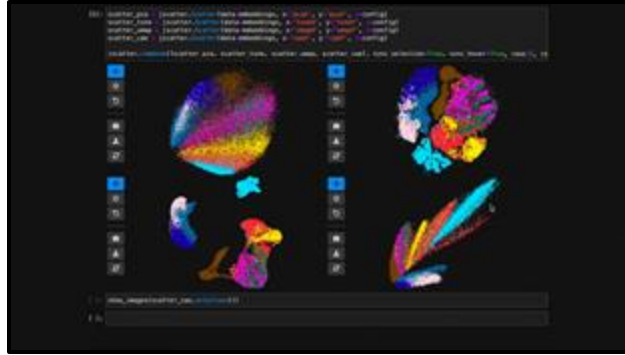
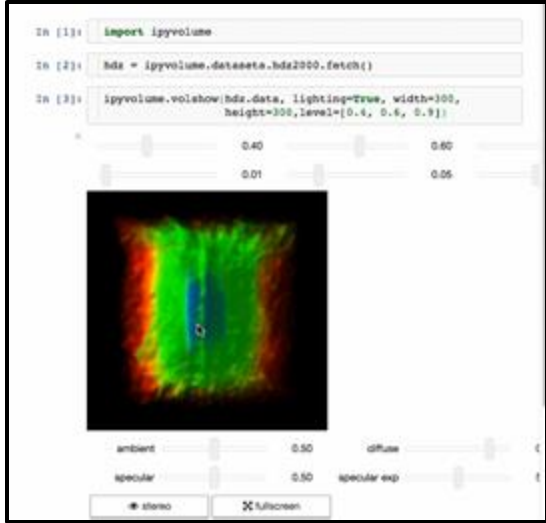
Vibrant ecosystem

Bridge library	Purpose	Key standard	Ecosystem
Anywidget 	Interactivity and visualization	ECMAScript Modules 	Jupyter Lab, VS Code, Colab, Marimo, Shiny, and more...
Oxbow 	Genomic data retrieval and manipulation	Apache Arrow 	PyArrow, Pandas, Polars, DuckDB, Dask, ...

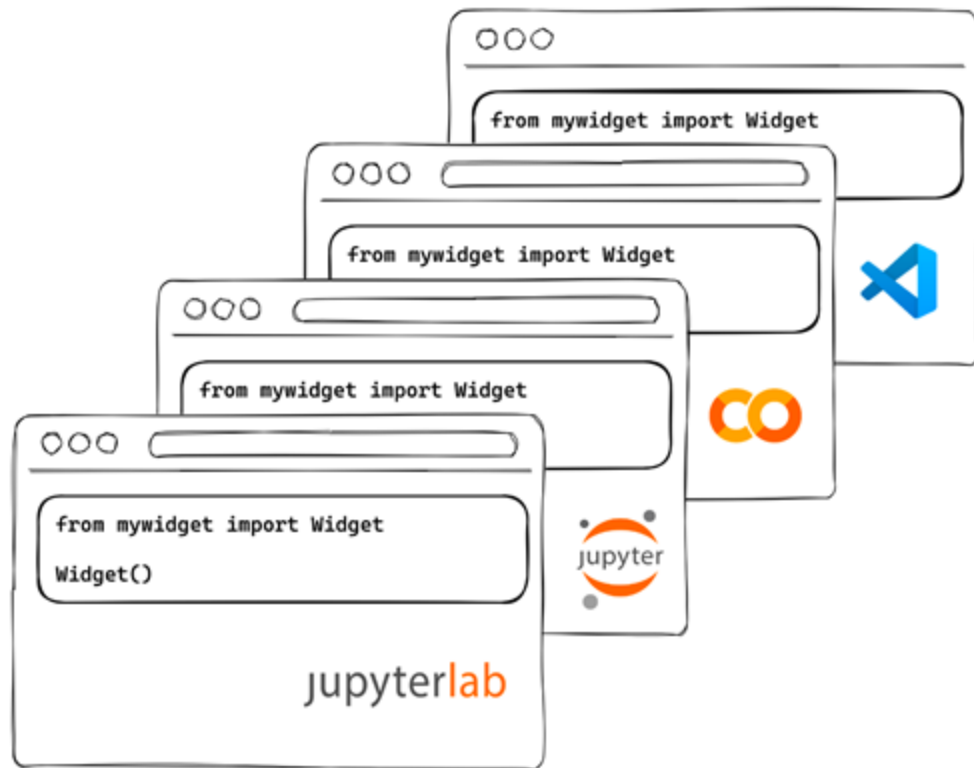
1. Interactivity and data visualization

Why Widgets









 **binder**

 **SOLARA**

 *jupyterlite*

 **Panel**

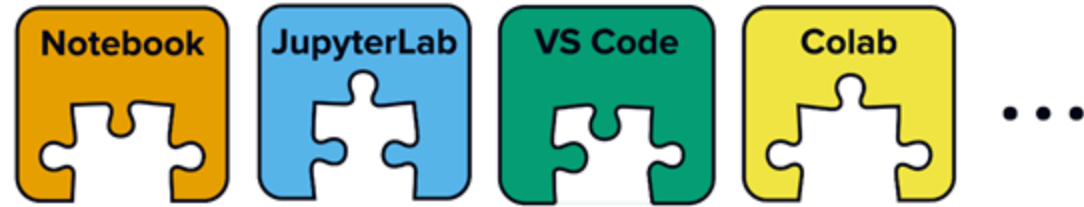
Shiny for Python



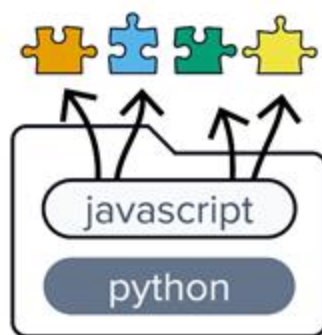
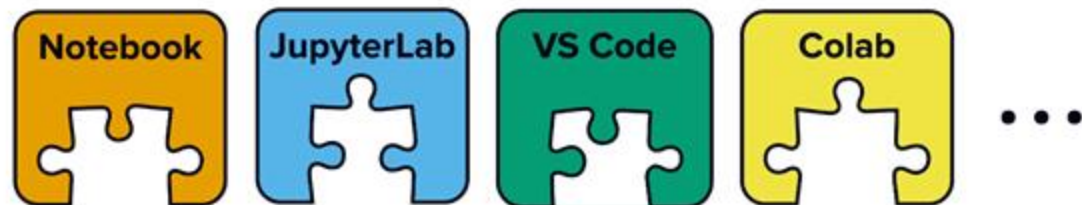
widget



Jupyter-Compatible Platforms (JCPs)



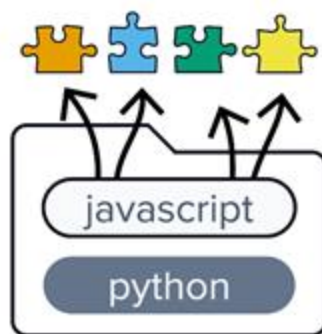
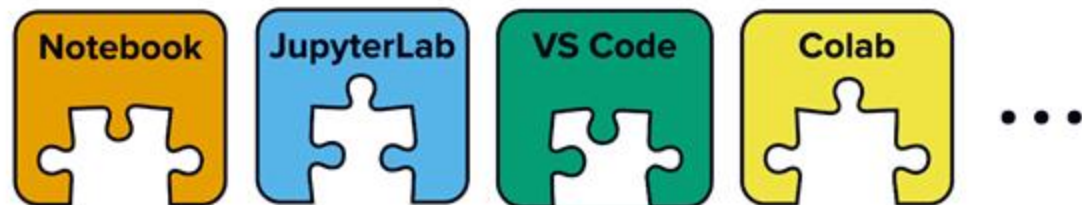
Jupyter-Compatible Platforms (JCPs)



widget

 JCP-specific front-end code

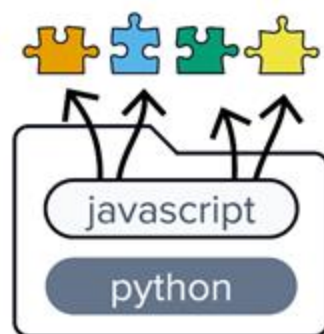
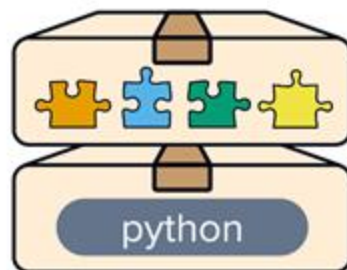
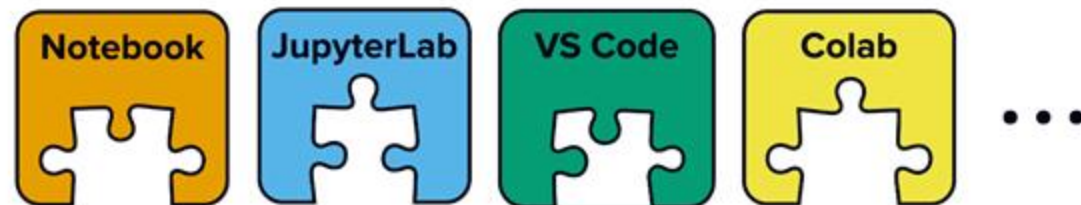
Jupyter-Compatible Platforms (JCPs)



 JCP-specific front-end code

widget

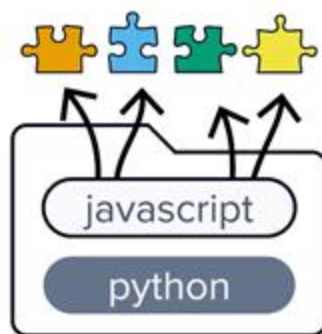
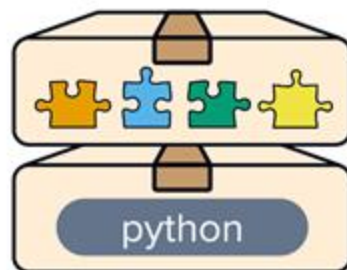
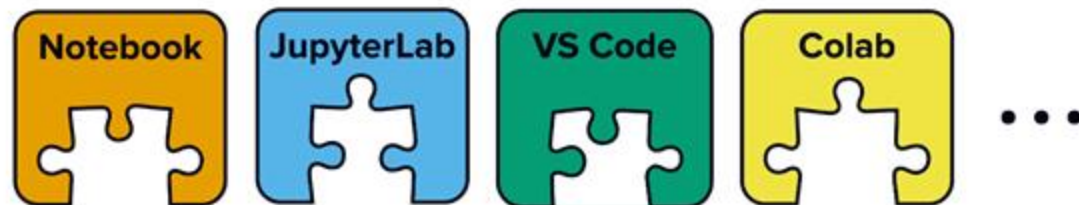
Jupyter-Compatible Platforms (JCPs)



widget

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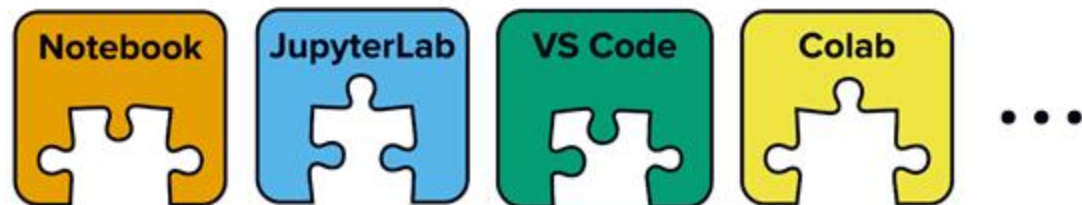
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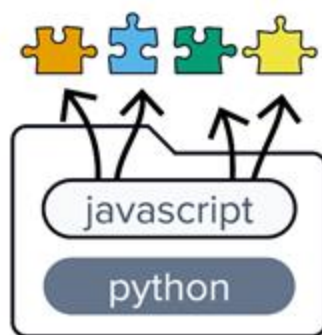
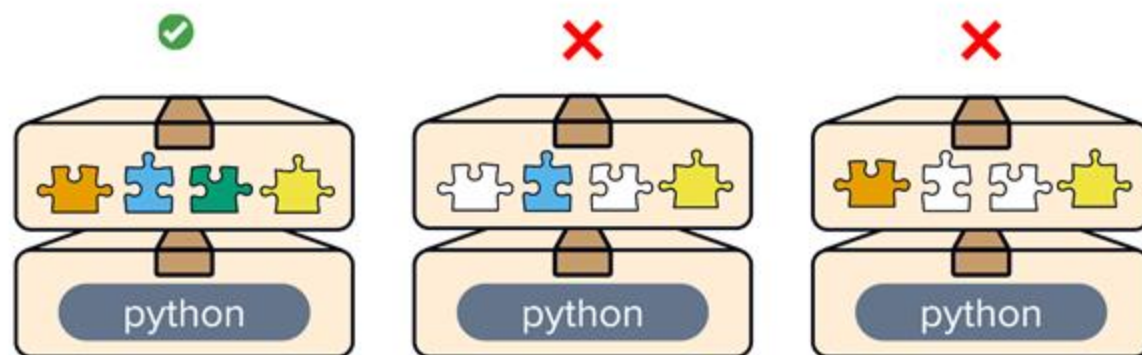
widget

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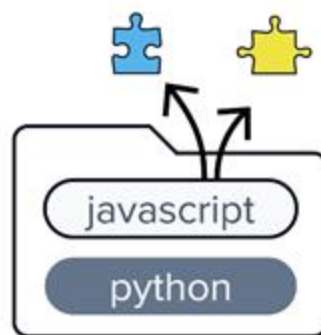


webserver

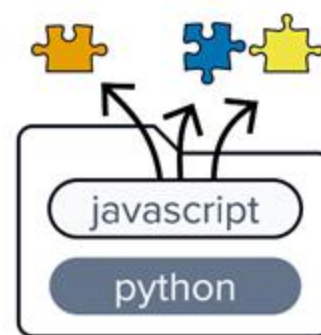
kernel



widget a



widget b



widget c

Jupyter-Compatible Platforms (JCPs)

Notebook

JupyterLab

VS Code

Colab

✓ No longer installable in google colab?

#204 by mccalluc was closed on Sep 6, 2022

✓ it stuck at "loading"

#67 by brainfo was closed on Aug 20, 2020

5

✓ nbconvert dependency issue

#152 by keller-mark was closed on May 9, 2022

✓ Jupyter Lab integration fails until display code is updated

#4 by flekschas was closed on Sep 4, 2019

1

⊙ Model not found error while running higlass

#86 opened on Dec 3, 2021 by sachingadakh

Installation issue

#8 by LeilyR was closed on Sep 4, 2019

✓ Interactive HiGlass view not displaying in Jupyter Notebook

#94 by MyBiscuit was closed on Jul 1, 2022

Support jupyterlab 3 ✗

#92 by alexlenail was merged on Mar 4, 2022 5 tasks

✓ react version conflict

#85 by kayakNIH was closed on Mar 4, 2022

1

🔗 Flekschas/add jupyter lab v2 support ✓

#64 by flekschas was merged on Jun 29, 2020 • 1 review approval 2 of 4 tasks

bug

enhancement

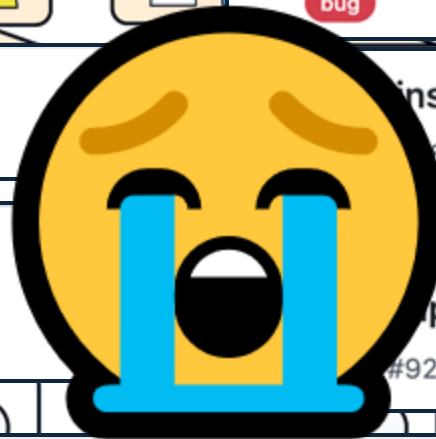
widget a

widget

✓ Jupyter Lab 2.1.5 fails to build after installing higlass-jupyter

1

5





anywidget

Custom Widgets Made Easy



 @manzt

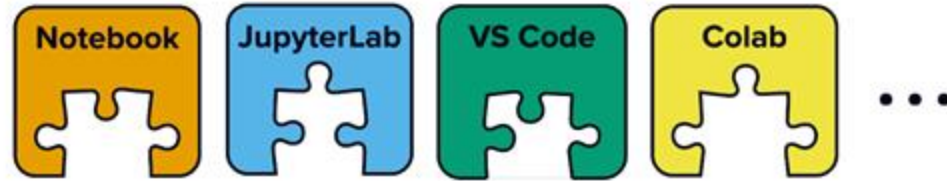


See also:

SciPy 2024, Tacoma, WA

Talk: anywidget: custom Jupyter Widgets made easy

Jupyter-Compatible Platforms (JCPs)

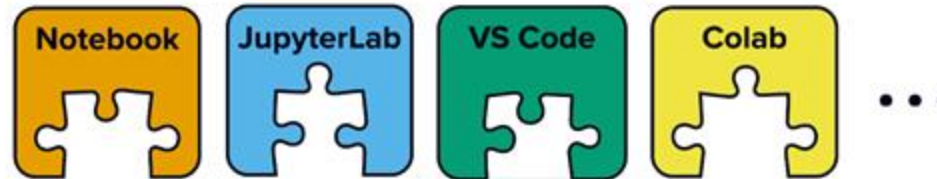


 **anywidget**



widget

Jupyter-Compatible Platforms (JCPs)



Search...

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- 11 ECMAScript Language: Source Text
- 12 ECMAScript Language: Lexical Gr...
- 13 ECMAScript Language: Expressions
- 14 ECMAScript Language: Statement...
- 15 ECMAScript Language: Functions ...
- 16 ECMAScript Language: Scripts an...
- 17 Error Handling and Language Ext...
- 18 ECMAScript Standard Built-in Obj...
- 19 The Global Object
- 20 Fundamental Objects
- 21 Numbers and Dates
- 22 Text Processing
- 23 Indexed Collections
- 24 Keyed Collections
- 25 Structured Data
- 26 Managing Memory
- 27 Control Abstraction Objects
- 28 Reflection
- 29 Memory Model

ECMA-262, 13th edition, June 2022 ECMAScript® 2022 Language Specification



About this Specification

The document at <https://tc39.es/ecma262/> is the most accurate and up-to-date ECMAScript specification. It contains the content of the most recent yearly snapshot plus any [finished proposals](#) (those that have reached Stage 4 in the proposal process and thus are implemented in several implementations and will be in the next practical revision) since that snapshot was taken.

This document is available as a [single page](#) and as [multiple pages](#).

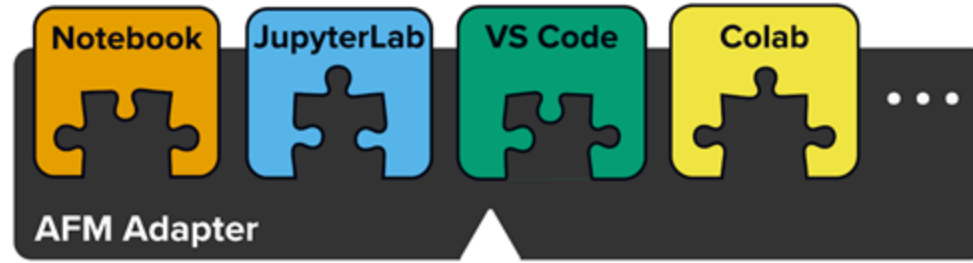


Anywidget
Front-end Module (AFM)



```
export default {  
  
  initialize({ model }) {  
    // initialize the widget model (optional)  
  },  
  
  render({ model, el }) {  
    // render a view of the model  
  },  
  
}
```

Jupyter-Compatible Platforms (JCPs)

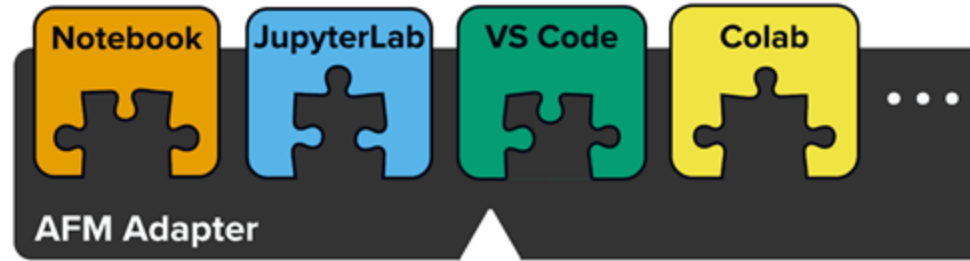


 **anywidget**



widget

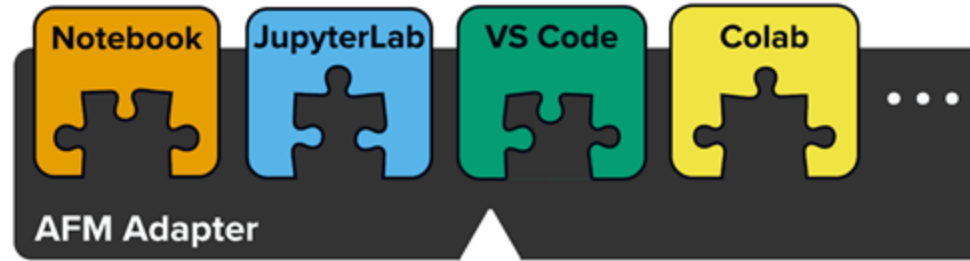
Jupyter-Compatible Platforms (JCPs)



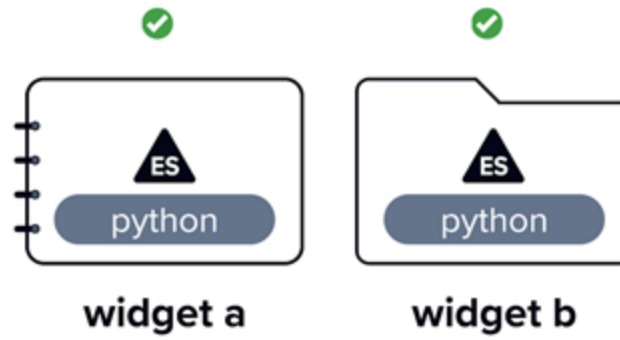
 **anywidget**



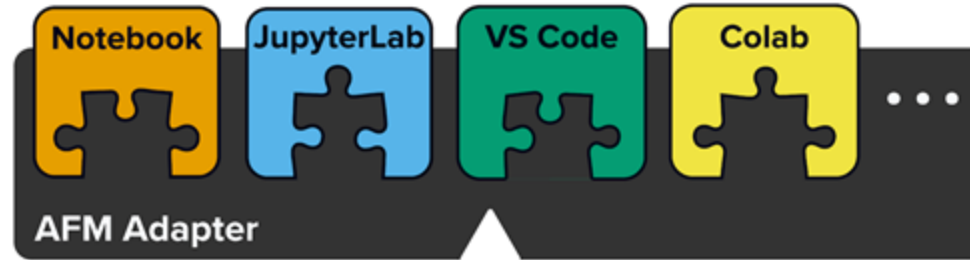
Jupyter-Compatible Platforms (JCPs)



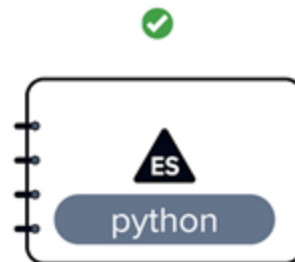
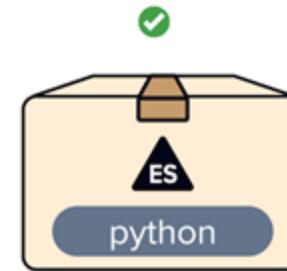
 **anywidget**



Jupyter-Compatible Platforms (JCPs)



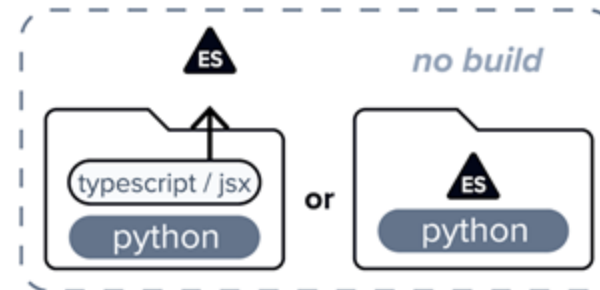
 **anywidget**



widget a

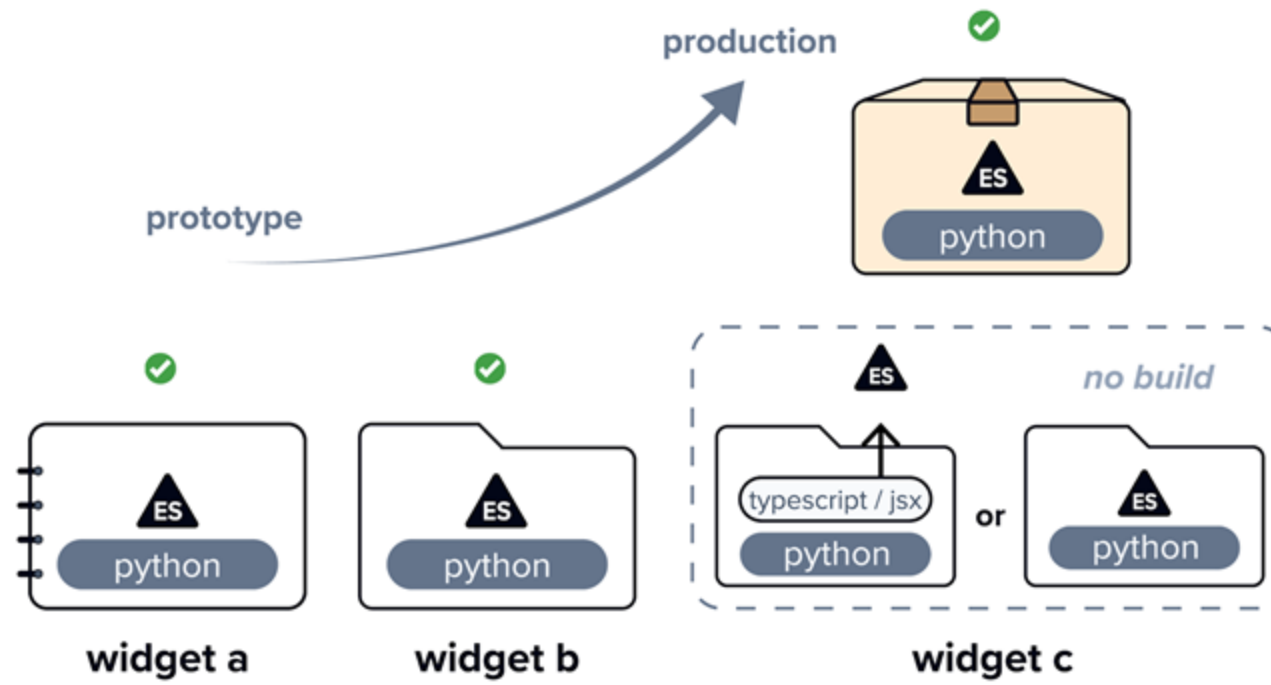
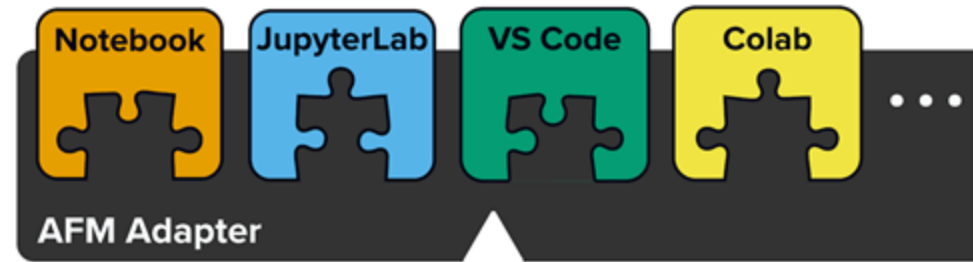


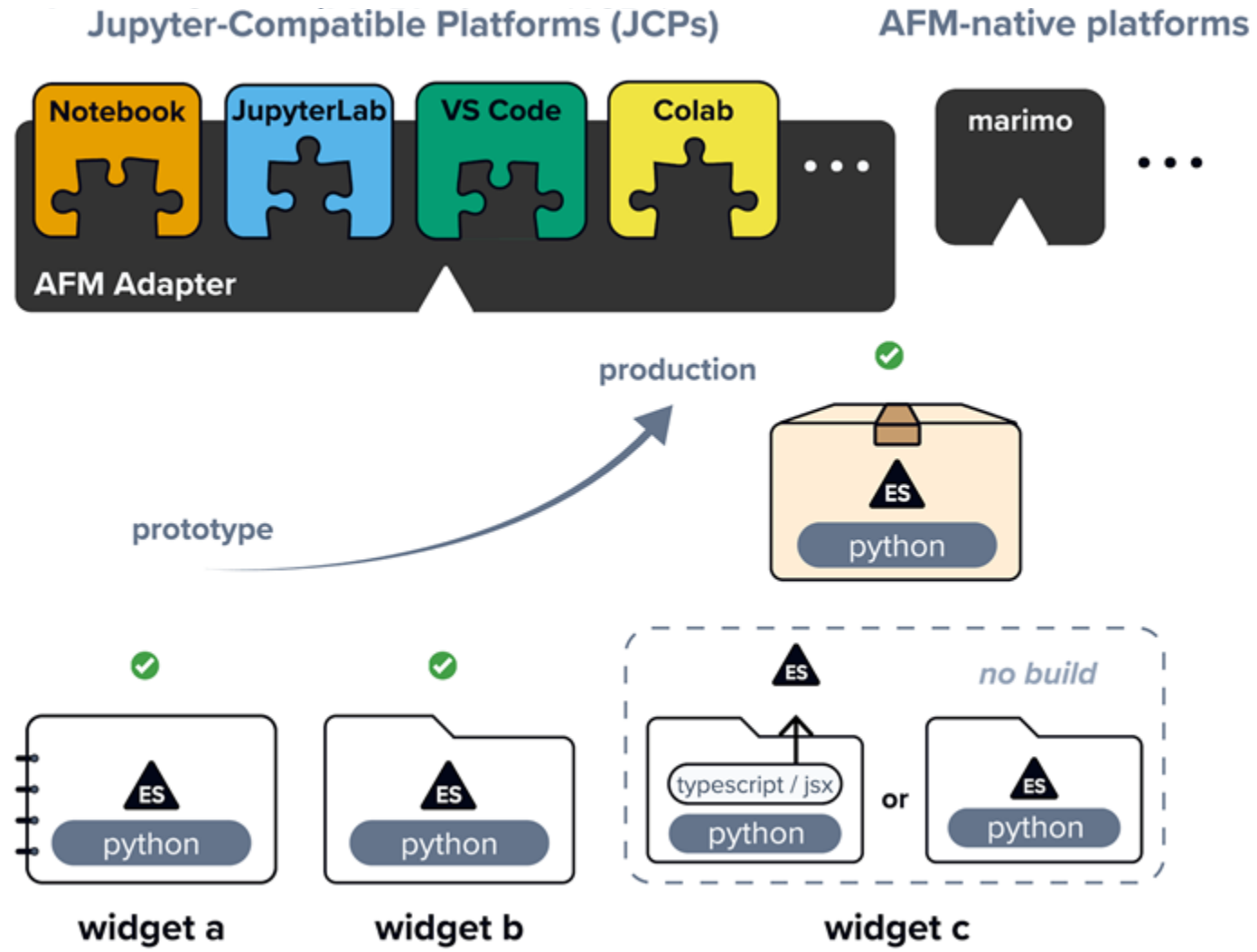
widget b



widget c

Jupyter-Compatible Platforms (JCPs)





Launcher × anywidget_example.ipynb +

Python 3 (ipykernel)

```
[ ]: import anywidget
import traitlets

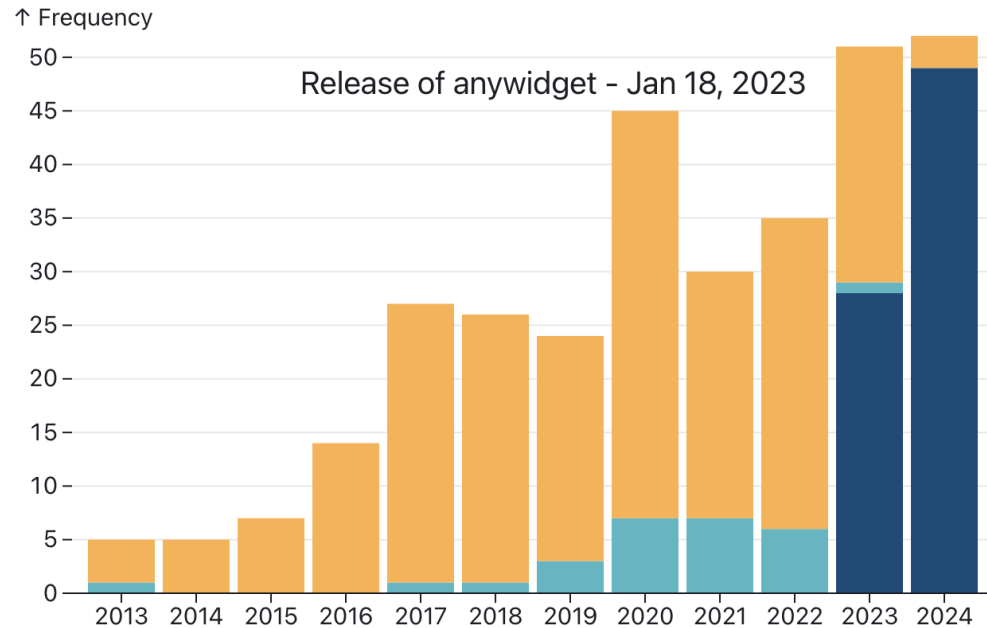
class CounterWidget(anywidget.AnyWidget):
    _esm = """
function render({ model, el }) {
  let count = () => model.get("value");
  let btn = document.createElement("button");
  btn.classList.add("counter-button");
  btn.innerHTML = `count is ${count()}`;
  btn.addEventListener("click", () => {
    model.set("value", count() + 1);
    model.save_changes();
  });
  model.on("change:value", () => {
    btn.innerHTML = `count is ${count()}`;
  });
  el.appendChild(btn);
}
export default { render };
"""
    _css = """
.counter-button {
  background-color: orange;
  border: 0;
  border-radius: 10px;
  padding: 10px 50px;
}
"""
    value = traitlets.Int(0).tag(sync=True)
```

```
[ ]: counter = CounterWidget()
counter
```

Impact

Custom Jupyter Widgets Per Year

■ created with anywidget ■ ported to anywidget ■ without anywidget



<https://github.com/manzt/anywidget-usage>

Manz, et al. (2024) Any notebook served: authoring and sharing reusable interactive widgets. <https://doi.org/10.25080/NRPV2311>

Manz, et al. (2024) anywidget: reusable widgets for interactive analysis and visualization in computational notebooks. <https://doi.org/10.21105/joss.06939>

```
pip install anywidget
```

Discord: <https://discord.gg/W5h4vPMbDQ>

Gallery: <https://anywidget.dev/en/community>

Community

Widgets Gallery

anywidget powers several open source projects. To feature your anywidget, please submit a PR.

- altair-viz/altair** (9.1k stars): Declarative statistical visualization library for Python. Includes a scatter plot visualization.
- rerun-io/rerun** (5.7k stars): Visualize streams of multimodal data. Fast, easy to use, and simple to integrate. Built in Rust using...
- koaning/drawdata** (742 stars): Draw datasets from within Jupyter. Includes a scatter plot visualization.
- uwdata/mosaic** (619 stars): An extensible framework for linking databases and interactive views. Includes a heatmap visualization.

2. Retrieval and manipulation of NGS data

“Bioinformatics scientist” spends 50% of day converting data



A Day in the Life

- 7:00 Arrive at the lab and begin interpreting or learning algorithms
- 9:00 Develop program to convert file format for input into data analysis program
- 11:00 Converting files from one format to another format;

Read research article while program running

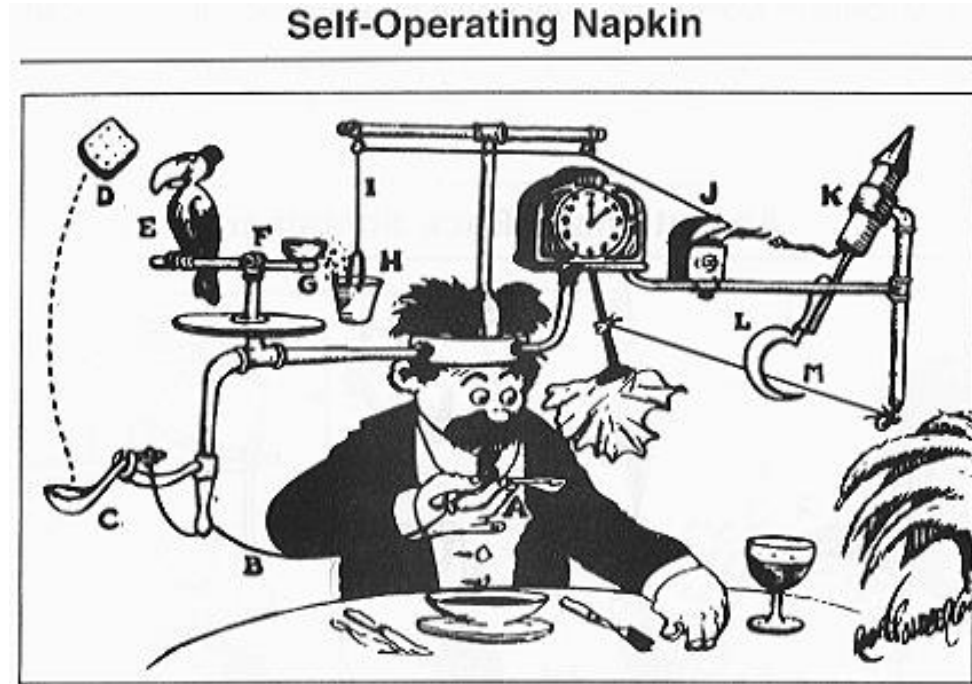
- 1:00 Input data into analysis program
- 1:30 Work on manuscript from a prior data analysis
- 2:00 Consult with other bioinformatic scientists or other medical professional

<https://www.decodingcancer.org/career/bioinformatics-scientist>

Rube Goldberg workflows

Lots of specialized file formats

FASTA, FASTQ, VCF/BCF, SAM/BAM, BED, GTF, ...



Unix Philosophy

Write programs that do one thing and do it well.

Write programs to work together.

Write programs to handle **text streams**, because that is a universal interface.

Curse of modularity

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Join nf-core

nf-core/modules

Browse the 1548 modules that are currently available as part of nf-core.

↓ Name

adapterremoval

Trim sequencing adapters and collapse overlapping reads

trimming adapters merging fastq

Included in: mag taxp profiler

adapterremovalfixprefix

Fixes prefixes from AdapterRemoval2 output to make sure no clashing read names are in the output. For use with DeDup.

adapterremoval fastq dedup

ariba_getref

Download and prepare database for Ariba analysis

fastq assembly resistance virulence

ariba_run

Query input FASTQs against Ariba formatted databases

fastq assembly resistance virulence

art_illumina

Simulation tool to generate synthetic Illumina next-generation sequencing reads

fastq fasta illumina simulate

Included in: readsimulator

artic_guppyplex

Aggregates fastq files with demultiplexed reads

artic aggregate demultiplexed reads

Included in: viralrecon

cat_fastq

Concatenates fastq files

cat fastq concatenate

Included in: airrflow bacass bamtofastq

and 26 more pipelines

Curse of modularity

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⊞ ≡

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and 26 more pipelines

Files

master

Go to file

- bwameth
- cadd
- cafe
- calder2
- canu
- cat
 - tests
 - environment.yml
 - main.nf
 - meta.yml
- catpack
- cdhit
- celesta
- cellbender
- cellpose
- cellranger
- cellrangerarc
- cellrangeratac
- cellsnr/modea
- centrifuge
- checkm
- checkm2
- checkqc
- checkv
- chewbbaca
- chopper
- chromap
- chromograph
- circexplorer2
- circularmapper
- clair3
- clame
- clipkit
- clippy
- clonalfameml
- clustalo
- cmseq/polymut
- cnvkit
- cnvnator
- cnvpytor
- cobrameta
- cobs

modules / modules / nf-core / cat / fastq / main.nf

nvnieuwk Fix issues with cat/fastq (#7517)

Code Blame 88 lines (78 loc) · 3.85 KB

```
1 process CAT_FASTQ {
2   tag "${meta.id}"
3   label 'process_single'
4
5   conda "${moduleDir}/environment.yml"
6   container "${workflow.containerEngine} == 'singularity' && !task.ext.singularity_pull_docker_container
7     ? 'https://community-cr-prod.sequera.io/docker/registry/v2/blobs/sha256/52/52ccce28d2ab928ab862e25aae26314d69c8e38bd41ca9431c67ef05221348aa/data'
8     : 'community.wave.sequera.io/library/coreutils_grep_gzip_lbzip2_pruned:838ba80435a629f8'"
9
10  input:
11    tuple val(meta), path(reads, stageAs: "input/*")
12
13  output:
14    tuple val(meta), path("*.merged.fastq.gz"), emit: reads
15    path "versions.yml", emit: versions
16
17  when:
18    task.ext.when == null || task.ext.when
19
20  script:
21    def prefix = task.ext.prefix ?: "${meta.id}"
22    def readList = reads instanceof List ? reads.collect { it.toString() } : [reads.toString()]
23    if (meta.single_end) {
24      if (readList.size >= 1) {
25        cat ${readList.join(' ')} > ${prefix}.merged.fastq.gz
26
27        cat <<-END_VERSIONS > versions.yml
28        "${task.process}":
29          cat: |${echo |${cat --version 2>&1} | sed 's/^.*coreutils/'; s/ .*$/'}
30        END_VERSIONS
31      } else {
32        error("Could not find any FASTQ files to concatenate in the process input")
33      }
34    } else {
35      if (readList.size >= 2) {
36        def read1 = []
37        def read2 = []
38        readList.eachWithIndex { v, ix -> (ix & 1 ? read1 : read2) << v }
39
40        cat ${read1.join(' ')} > ${prefix}_1.merged.fastq.gz
41        cat ${read2.join(' ')} > ${prefix}_2.merged.fastq.gz
42
43        cat <<-END_VERSIONS > versions.yml
44        "${task.process}":
45          cat: |${echo |${cat --version 2>&1} | sed 's/^.*coreutils/'; s/ .*$/'}
46        END_VERSIONS
47      } else {
48        error("Could not find any FASTQ file pairs to concatenate in the process input")
49      }
50    }
51  }
52
53  stub:
54    def prefix = task.ext.prefix ?: "${meta.id}"
55    def readList = reads instanceof List ? reads.collect { it.toString() } : [reads.toString()]
56    if (meta.single_end) {
57      if (readList.size >= 1) {
58        echo '' | gzip > ${prefix}.merged.fastq.gz
59
60        cat <<-END_VERSIONS > versions.yml
61        "${task.process}":
62          cat: |${echo |${cat --version 2>&1} | sed 's/^.*coreutils/'; s/ .*$/'}
63        END_VERSIONS
64      } else {
65        error("Could not find any FASTQ files to concatenate in the process input")
66      }
67    }
68  }
69
70  if (readList.size >= 2) {
71    echo '' | gzip > ${prefix}_1.merged.fastq.gz
72
73    cat <<-END_VERSIONS > versions.yml
74    "${task.process}":
75      cat: |${echo |${cat --version 2>&1} | sed 's/^.*coreutils/'; s/ .*$/'}
76    END_VERSIONS
77  }
78  }
```

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Simulation tool to generate synthetic Illumina next-generation sequencing reads

fastq fasta illumina simulate

Included in: readsimulator

artic_guppyplex

Aggregates fastq files with demultiplexed reads

artic aggregate demultiplexed reads

Included in: viralrecon

cat_fastq

Concatenates fastq files

cat fastq concatenate

Included in: airflow bacass bamtofastq

and 26 more pipelines



Files

modules / modules / nf-core / cat / fastq / main.nf

master

nvnieuwk Fix issues with cat/fastq (#7517)

npm

find packages

sign up or log in

left-pad

public

String left pad



npm install left-pad

azer published 4 months ago

0.0.4 is the latest of 5 releases

github.com/azer/left-pad

BSD license

Collaborators



Stats

45,134 downloads in the last day

320,537 downloads in the last w...

1,587,516 downloads in the last ...

No open issues on GitHub

One open pull request on GitHub

Install

```
$ npm install left-pad
```

Usage

```
leftpad = require('left-pad')

leftpad('foo', 5)
// => "  foo"

leftpad('foobar', 6)
// => "foobar"

leftpad(1, 2, 0)
// => "01"
```

Cross-Language Apps
& Web Services

Exploratory Analysis
& AI/ML

Cloud Computing
& Data Pipelines

Tools **tightly coupled to format I/O**
Rely on **low-level** reference libraries for manipulation

Research 🤔



Development 🤔



CRAM

SAM

BAM

VCF

BCF

FASTQ

FASTA

Two fundamental data modalities

Table-ish

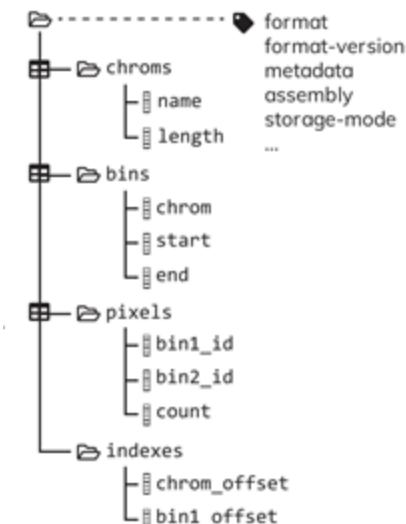
Col	Field	Type	Regexp/Range	Brief description
1	QNAME	String	[!~?A-Z]{1,254}	Query template NAME
2	FLAG	Int	[0, 2 ¹⁶ - 1]	bitwise FLAG
3	RNAME	String	* [:rname:^*]*[:rname:]*	Reference sequence NAME ¹¹
4	POS	Int	[0, 2 ³¹ - 1]	1-based leftmost mapping POSition
5	MAPQ	Int	[0, 2 ⁸ - 1]	MAPping Quality
6	CIGAR	String	* ([0-9]+[MIDNSHPX=])+	CIGAR string
7	RNEXT	String	* [:rname:^*]*[:rname:]*	Reference name of the mate/next read
8	PNEXT	Int	[0, 2 ³¹ - 1]	Position of the mate/next read
9	TLEN	Int	[-2 ³¹ + 1, 2 ³¹ - 1]	observed Template LENgth
10	SEQ	String	* [A-Za-z=.] +	segment SEQUENCE
11	QUAL	String	[!~]+	ASCII of Phred-scaled base QUALity+33

htslib/bgzf
SAM (BAM)
VCF (BCF)

multiscale binary
bigWig
bigBed

TSV
BedGraph
BED
GFF/GTF
...
Many more!

Array-ish



AnnData (HDF5,
Zarr)

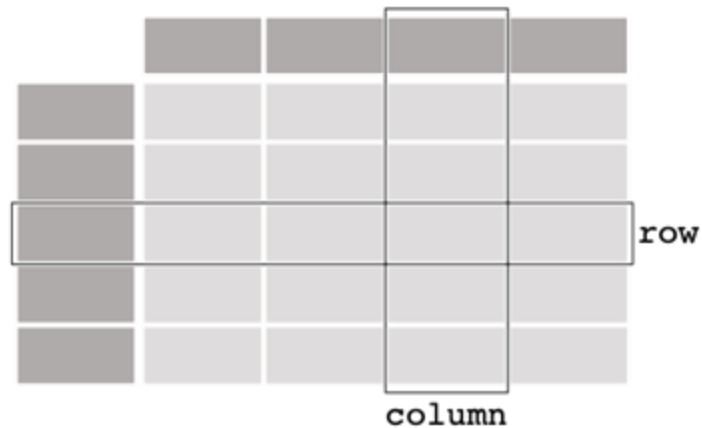
Cooler (HDF5)

TIFF

.hic (custom binary)

Focus on the *representation*, not the storage format

Tabular → DataFrame



Array-based → NDAarray

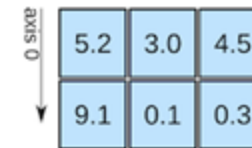
1D array



axis 0

shape: (4,)

2D array

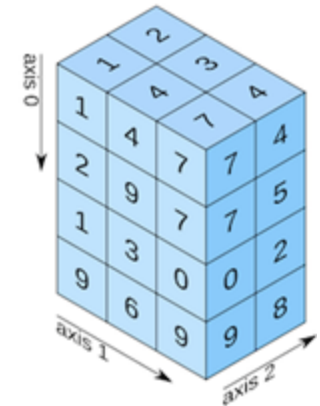


axis 0

axis 1

shape: (2, 3)

3D array



axis 0

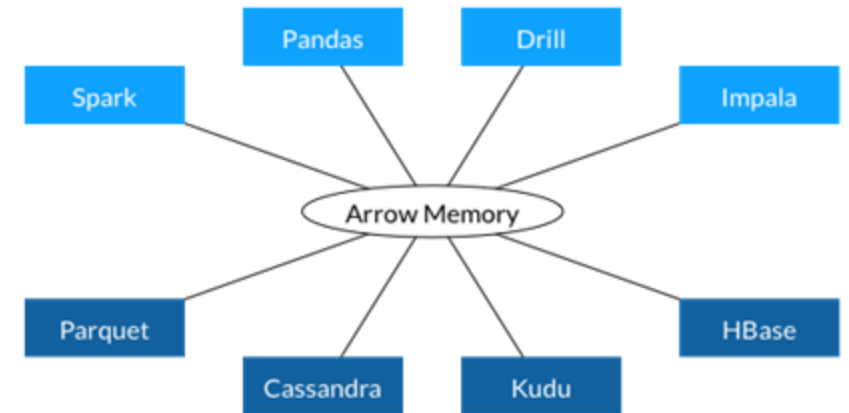
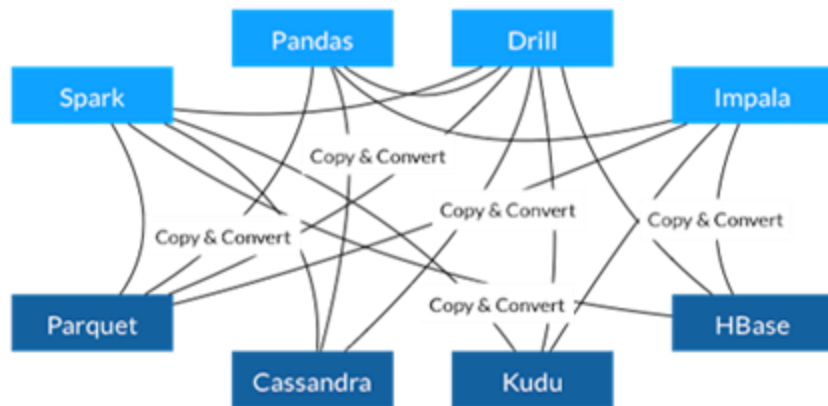
axis 1

axis 2

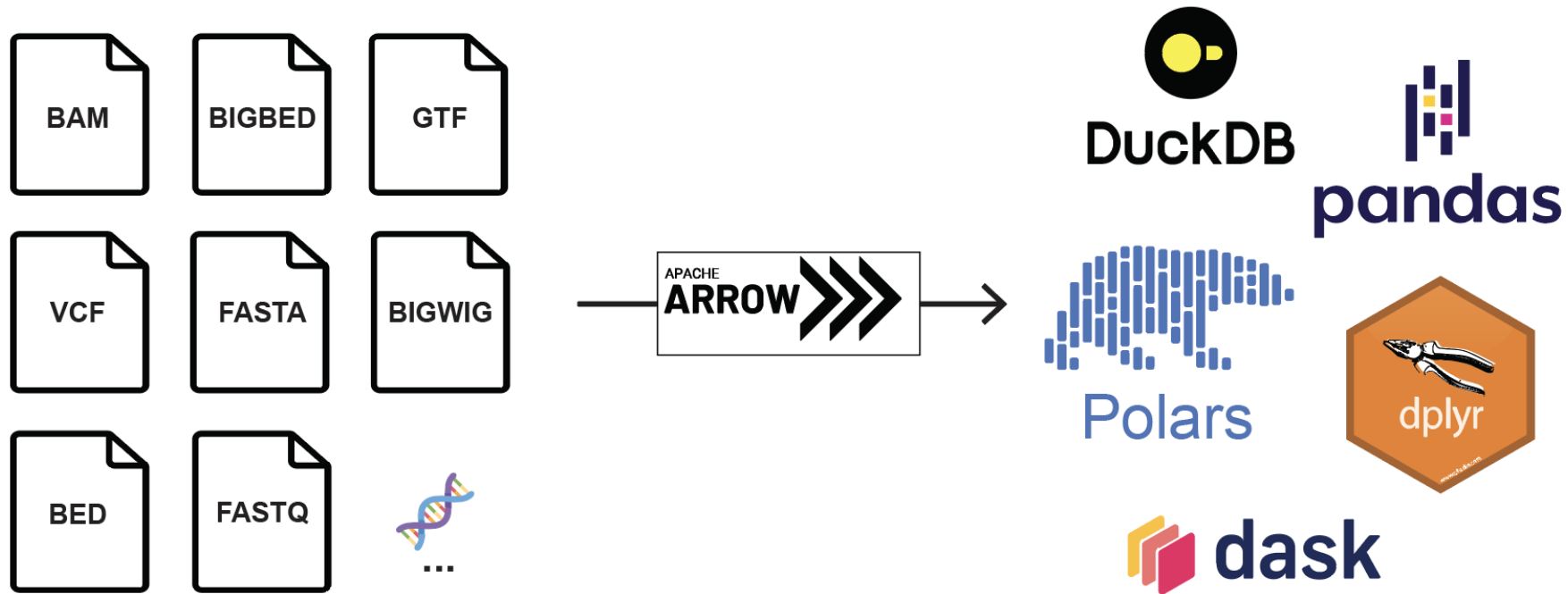
shape: (4, 3, 2)

A standard representation: Apache Arrow

- The file format “problem” is not unique to bioinformatics
- Arrow is a unified **columnar, in-memory layout** for tabular data
- Organized for efficient analytic operations on CPU and GPU
- Supports zero-copy reads for fast access and IPC



oxbow



file access 

Noodles: <https://github.com/zaeleus/noodles>

Bigtools: <https://github.com/jackh726/bigtools>

Bioinformatics, 2024, **40**(6), btae350
<https://doi.org/10.1093/bioinformatics/btae350>
Advance Access Publication Date: 5 June 2024
Applications Note

OXFORD

Data and text mining

Bigtools: a high-performance BigWig and BigBed library in Rust

Jack D. Huey^{1,2,*} and Nezar Abdennur^{3,4,*}



Jack Huey

It's just a DataFrame!

The screenshot shows a Jupyter Notebook interface with a file explorer on the left and a code editor on the right. The code editor contains three cells of Python code. The first cell imports `oxbow` as `ox` and creates a `ds` object from a BAM file. The second cell converts `ds` to a pandas DataFrame `df`. The third cell displays the first 10 rows of `df` using `df['tags'].struct.unnest()`.

```
[1]: import oxbow as ox
ds = ox.from_bam("data/sample.bam")

[2]: df = ds.pl()
df
[2]: shape: (10, 13)
```

	qname	flag	rname	pos	mapq	cigar	rnext	pnext	tlen	seq	qual	end
	str	u16	cat	i32	u8	str	cat	i32	i32	str	str	i32
	"HWI-BRUNOP16X_0001:3:48:4861:1..."	163	"chr1"	10542	0	"50M"	"chr1"	10571	79	"CGAAATCTGTGCAGAGGAGAACGCAGCTCC..."	"99999999999999999999999999999999..."	10591
	"HWI-BRUNOP16X_0001:3:28:6650:1..."	16	"chr1"	10546	16	"75M"	null	null	0	"ATCTGTGCAGAGGAGAACGCAGCTCCGCC..."	"fgggggggdgdggcdgfggggfgggggg..."	10620
	"HWI-BRUNOP16X_0001:3:8:20066:8..."	16	"chr1"	946457	0	"75M"	null	null	0	"TAGTCCGAGGTCTCTCTGAACCTTCCCAAGC..."	"BBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBB..."	946531
	"HWI-BRUNOP16X_0001:3:27:10302:..."	16	"chr1"	1014060	37	"75M"	null	null	0	"AGCTGAATGGGCAGGTCCCCAGAAGATCG..."	"BBBBBBBBBBBBBBBBBcYRcYfgfgf_gf..."	1014134
	"HWI-BRUNOP16X_0001:3:65:3144:1..."	83	"chr3"	196957	60	"50M"	"chr3"	196008	-999	"GTAACGCTCCCGACCTGCGCGCCCCCGT..."	"BBBBBBBBBBBBBBB_TTSSS[[Obbd`]e^..."	197006
	"HWI-BRUNOP16X_0001:3:68:13088:..."	16	"chr3"	196958	37	"75M"	null	null	0	"GACCCCCCGGCCCCCGCGCCCCCGCC..."	"BBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBB..."	197032
	"HWI-BRUNOP16X_0001:3:48:3417:1..."	163	"chr3"	196961	60	"50M"	"chr3"	319702	122791	"GCTTACCGGACCTGCGCGCCCCCGTCCG..."	"gggggggggggggggggggggggfdagggg..."	197010
	"HWI-BRUNOP16X_0001:3:46:17583:..."	161	"chrX"	503847	0	"50M"	"chr4"	185365552	0	"TTTTATTTTTTTTTTGAGTGGAGTCTCG..."	"ddfdfd____dffj__aeZ]XZSPSNS..."	503896
	"HWI-BRUNOP16X_0001:3:4:7989:14..."	16	"chrY"	586185	0	"75M"	null	null	0	"GTGCGATCTCGGTCGCTGCAACCTCTGCT..."	"BBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBB..."	586259
	"HWI-BRUNOP16X_0001:3:44:11450:..."	0	"chrY"	587561	0	"75M"	null	null	0	"NNTGCAGTGAGCTGAGATTGTGCCACTGCA..."	"BBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBB..."	587635

```
[3]: df['tags'].struct.unnest()
[3]: shape: (10, 12)
```

AM	MD	NM	RG	SM	X0	X1	XA	XG	XM	XO	XT
i64	str	i64	str	i64	i64	i64	str	i64	i64	i64	str
0	"18C31"	1	"brain_50_fcb"	0	3	8	null	0	1	0	"82"
null	"14C52A7"	2	"brain_75_fca"	null	1	5	null	0	2	0	"85"
null	"2T0G5T65"	3	"brain_75_fca"	null	2	0	"2,-131443143,75M,3;"	0	3	0	"82"
null	"7G1C4A2A57"	4	"brain_75_fca"	null	1	0	null	0	4	0	"85"
37	"OC0A0G1G0C0T1A41"	7	"brain_50_fcb"	37	1	0	null	0	7	0	"85"
null	"0A0G0A0G1T0T0A1C1G0A3T4G6T4G1T..."	19	"brain_75_fca"	null	1	0	null	0	19	0	"85"
37	"50"	0	"brain_50_fcb"	37	1	0	null	0	0	0	"85"
0	"4T36C8"	2	"brain_50_fcb"	0	18	174	null	0	2	0	"82"

Simple 2 Python 3 (ipykernel) | Idle Saving completed Mode: Command Ln 1, Col 1 demo.ipynb 0



Name	Modified
data	6d ago
pyproject.toml	6d ago
quickstart.ipynb	1s ago
uv.lock	6d ago

quickstart.ipynb

+ ✂ 📄 📌 ▶ ⏮ ⏭ Markdown ▾

Python 3 (ipykernel) ⌵ ☰

pip install oxbow

Oxbow lets you access potentially larger-than-memory genomic files as tabular data structures, such as data frames.

Create a DataSource

Use the convenience function associated with your file type. The returned `DataSource` object can be used to access the data in the file.

```
[ ]: import oxbow as ox
import polars as pl

ds = ox.from_bam("data/sample.bam")
```

Into data frames

If the dataset fits comfortably in memory, you can materialize it fully as a [Pandas](#) or [Polars](#) data frame.

```
[ ]: ds.pd() # or ds.to_pandas()
```

```
[ ]: ds.pl() # or ds.to_polars()
```

Into lazy data structures

If the data source is very large, you can also load it into a lazy or "out-of-core" data structure, such as a Polars lazy frame or [Dask](#) data frame.

```
[ ]: import polars as pl

df = ds.pl(lazy=True)
df.show_graph()
```

It can be a distributed dataframe...

File Edit View Run Kernel Tabs Settings Help

http://127.0.0.1:8787

AGGREGATE TIME PER ACTION

BANDWIDTH TYPES

BANDWIDTH WORKERS

CLUSTER MAP

CLUSTER MEMORY

COMPUTE TIME PER KEY

CONTENTION

CPU

EXCEPTIONS

FINE PERFORMANCE METRICS

GPU MEMORY

GPU UTILIZATION

GRAPH

GROUP PROGRESS

GROUPS

MEMORY BY KEY

NPROCESSING

OCCUPANCY

PROFILE

PROFILE SERVER

PROGRESS

RMM MEMORY

SCHEDULER SYSTEM

TASK STREAM

WORKERS

WORKERS CPU TIMESERIES

WORKERS DISK

WORKERS DISK TIMESERIES

WORKERS MEMORY

WORKERS MEMORY TIMESERIES

WORKERS NETWORK

WORKERS NETWORK TIMESERIES

WORKERS TRANSFER BYTES

oxbow-demo-v3.ipynb

Python 3 (ipykernel)

Oxbow and Dask

```
[1]: import os

import matplotlib.pyplot as plt
import numpy as np
import pandas as pd
import bioframe

from dask.distributed import LocalCluster, Client
import dask.dataframe as dd
import dask.array as da
import dask

import oxbow as ox
```

Start a Dask cluster

```
[2]: client = Client()
      client
```

```
[2]: Client
      Client-e079b63c-5b59-11f0-8812-0b9aefa8268b
      Connection method: Cluster object
      Cluster type: distributed.LocalCluster
      Dashboard: http://127.0.0.1:8787/status
      Launch dashboard in JupyterLab
```

Cluster Info

Load a big BAM file into a distributed DataFrame!

```
[ ]: file_path = (
      '/home/nezar/local/data/H1-H3K9me2_R1.target.markdup.sorted.bam'
      )

print(f"BAM file is {os.path.getsize(file_path) / 1024 / 1024 / 1024:.2f}G")

[ ]: chromsizes = bioframe.fetch_chromsizes('hg38')
      chromosomes = list(chromsizes.keys())
      bins = bioframe.binnify(chromsizes, 100_000_000)
      regions = (bins["chrom"] + ":" + (bins["start"] + 1).astype(str) + "-" + bins["end"].astype(str)).tolist()
```

Graph

Task Graph

Scheduler is empty.

Task Stream

Task Stream

206400us 206420us 206440us 206460us 206480us

06:38:30

Jan 21, 1970

Workers Memory

Bytes stored per worker

0.0 4.0 GiB 8.0 GiB 12.0 GiB 16.0 GiB 20.0 GiB 24.0 GiB 28.0 GiB

Simple 1 Python 3 (ipykernel) | Idle Mode: Command Ln 1, Col 1 oxbow-demo-v3.ipynb 1



pip install oxbow

</> github.com/abdenlab/oxbow

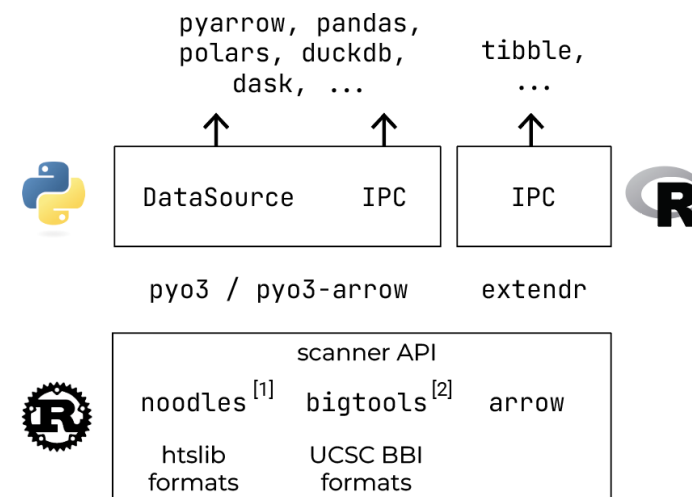


oxbow.readthedocs.io



oxbow.zulipchat.com

Format	Modality	Encoding	Zooms	Compression support	Index support	
FASTQ	Sequence	Text	No	BGZF		✓
FASTA	Sequence	Text	No	BGZF	FAI, FAI + GZI	✓
SAM	Alignment	Text	No	BGZF	TBI, CSI	✓
BAM	Alignment	Binary	No	BGZF	BAI	✓
CRAM	Alignment	Binary	No	Multiple	CRAI	Coming soon!
VCF	Variant call	Text	No	BGZF	TBI, CSI	✓
BCF	Variant call	Binary	No	BGZF	CSI	✓
GTF	Feature	Text	No	BGZF	TBI, CSI	✓
GFF3	Feature	Text	No	BGZF	TBI, CSI	✓
BED	Feature	Text	No	BGZF	TBI, CSI	✓
BigBed	Feature	Binary	Yes	deflate	Internal index	✓
BigWig	Feature	Binary	Yes	deflate	Internal index	✓



DataSource objects expose iterators of Arrow Record Batches.



Nezar Abdennur

Join us at Sprints!



Conrad Bzura

3. Putting it together


```
pip install higlass-python
```

796 Cancer sample ATAC-seq tracks

```
import bioframe
from hgillness_scale import Scale
hg38 = bioframe.fetch_chromsizes('hg38')
scale = Scale(hg38)

widget.zoom_to(
    viewconf.views[0].uid,
    int(scale(("chr8", 127_500_000))),
    int(scale(("chr8", 128_500_000)))
)
```

TCGA cCRE embedding by NMF

```
[ ]: import ipywidgets as widgets
from IPython.display import display
from nmf_vis.scatter import create_scatterplot
from nmf_vis.heatmap import create_heatmap
```

Explore cancer cCRE signatures

```
pip install higlass-python
```

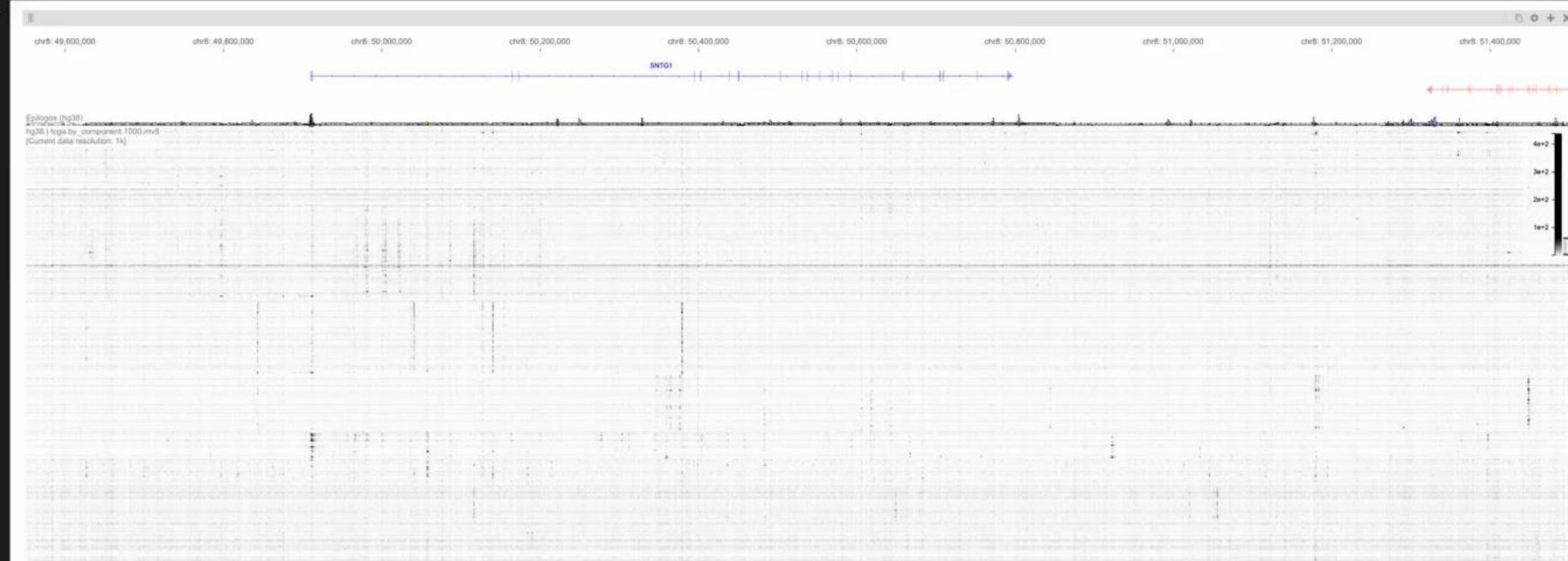
796 Cancer sample ATAC-seq tracks

```
import json
import higlass as hg

with open("conf/viewconf.json", "r") as f:
    viewconf = hg.Viewconf(**json.load(f))

widget = viewconf.widget()
widget
```

[4] ✓ 0.0s



PROBLEMS OUTPUT TERMINAL PORTS JUPYTER DEBUG CONSOLE

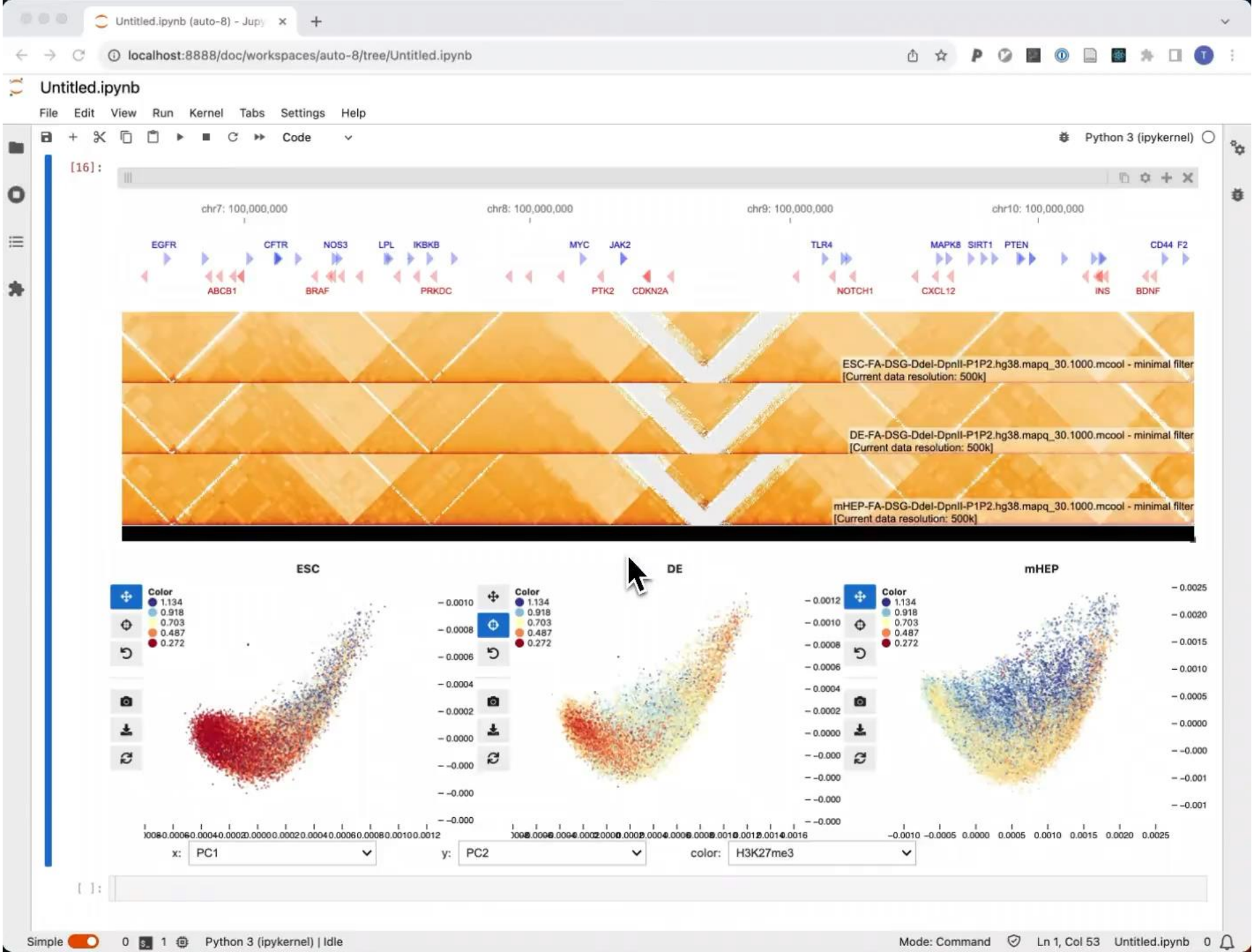
```
nmfvis
tcga-nmf on main [??] is v0.1.0 via v3.11.10 (nmfvis)
>
```

OUTLINE

TIMELINE

main* 0 0 0 Connect Live Share

Spaces: 4 LF () Cell 3 of 14



Composability and separation of concerns

File I/O	Data manipulation
Data storage	Memory layout
Data Transport	Decoding
Platform	Plugin

Composability and separation of concerns

File I/O
Data storage
Data Transport
Platform

Open
Standards



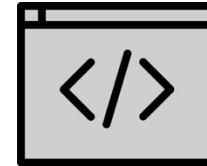
Data manipulation
Memory layout
Decoding
Plugin

Acknowledgments



Oxbow and Anywidget contributors

- Trevor Manz
- Conrad Bzura
- Jack Huey
- Vedat Yilmaz
- Garrett Ng
- Alejandro Gonzales-Irribarren
- Devin de Hueck



<https://github.com/abdenlab/oxbow>

<https://github.com/manzt/anywidget>



Other contributors

- Peter Kerpedjiev
- Fritz Lekschas
- HiGlass team
- Open2C Community
- HIDIVE (Gehlenborg) lab, HMS

<https://github.com/jackh726/bigtools>

<https://github.com/higlass/higlass-python>

<https://github.com/abdenlab/tcga-db>

<https://github.com/abdenlab/tcga-nmf>



ITCR contributors

- Nicole Shedd
- Anisha Mhatre
- Thomas Reimonn
- Zeeshan Siddiqui
- Weng and Moore labs, UMass Chan



Jupyter Scatter



HiGlass

